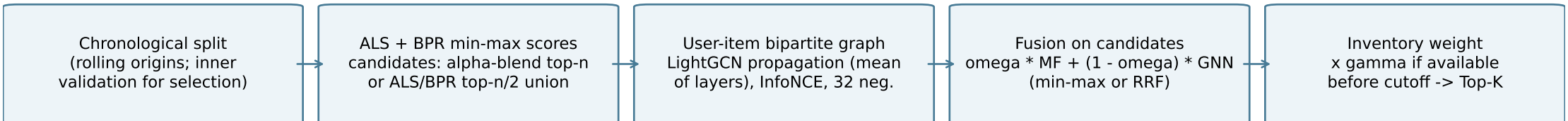

Pair weight w = sum of view 1 / cart 6 / purchase 12 events, each decayed with a 7-day half-life; ALS confidence 1 + alpha * w.
BPR and GNN InfoNCE loss weight w / mean(w) (BPR: negatives by item user count; GNN: unweighted propagation, edge dropout 0.1, 32 uniform negatives).

No Isolation Forest in the final model. Tuned parameters chosen on inner validation, fixed settings frozen in the plan and code hash;
final models refit with fixed epochs on the full pre-test window.